

Lap Simulation Parts;

- car.py
 - Handles most of the highest level logic
 - Most of the parts of the simulation in terms of things that the car does are handled by this part of the script.
 - References most of the other scripts
 - Generates for example the relevant tire model itself
- tires.py
 - One of the most important single parts of the car
 - Inherently handles all of the tire-specific calculations
 - Divided into physical tire and magic formula tire:
- motor.py
 - Handles the actual input/output of determining if a given torque can be provided
 - Uses the motor power curves to determine whether a specific torque (given RPM) is valid, and then determines what the operating characteristics will thus be.
 - Will have functions to limit torque based on the transient heat parameters
- track.py
 - Class to create and use tracks
 - Has visualization methods
- simulation.py
 - Combines all of the previous scripts to get the movement of the car around the track.
 - Maybe consider dividing into different simulation types?

Simulation

Simulation Parts

- Lateral Handling ;
 - This will be handled by an “optimizing” function.
- Longitudinal Handling ;
 - Based on where on the track the vehicle is and the *precalculated* velocity profile, we can determine whether we want to accelerate or decelerate.

Optimizing Cornering

1. Find the longest straight of the track
2. In reverse, iterate the track segments
3. For each track segment, determine input params for the following cases:
 - Maximum entry speed (what is the fastest speed that a car can enter the turn at)
 - Minimum segment time
4. Use the previous segments maximum entry speed as the maximum exit speed for the current segment
5. Always optimize for fastest

Idea;

determine the points myself. with some amount of calculation based on logic.

I set an entrance speed, maybe desired acceleration?

Current tasks

- ☐ Finish building out vehicle simulation
 - ☐ Implement Steering
 - ☐ Implement Throttle
 - ☐ Implement overall physics
- []

Physical Model

Physical Tire

tire_class

Holds all non-magic formula tire parameters and methods divided into the following categories:

Static parameters

- width
- diameter
- rotational inertia

Transient parameters

- steering angle
- angular velocity
- angular acceleration
- slip angle?

Methods

Magic Formula Tire

new_magic_formula_tire

Holds all the parameters and methods to do with calculating tire forces themselves.

Static parameters

- Magic tire formula parameters

Transient parameters

- longitudinal tire forces
- lateral tire forces

Methods